

LONGEST SUBSTRING WITHOUT REPEATING CHARACTERS

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OBJECTIVE / AIM

To write a Java program to find the longest substring without repeating characters using only String operations.

PROBLEM STATEMENT

Write a Java program to find the longest substring without repeating characters. Input: **abcabcbb** → Output: **3**, String: **abc**. Input: **pwwkew** → Output: **3**, String: **wke**. Note: **pwke** is not a substring; it is a subsequence.

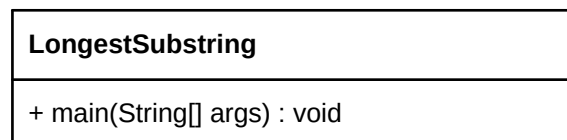
ALGORITHM / LOGIC

1. Read the input string.
2. Maintain two String variables: current and longest.
3. Get each character as a one-character String using substring().
4. Use indexOf() to check whether the character is already present in current.
5. If repeated, use substring() to remove the part through the previous occurrence.
6. Append the current character to current.
7. Update longest whenever current becomes longer.
8. Display longest and its length.

STEPS FOR EXECUTION

1. Save the program as **LongestSubstring.java**.
2. Compile: **javac LongestSubstring.java**
3. Run: **java LongestSubstring**
4. Enter the input string and observe the result.

UML CLASS DIAGRAM



SOURCE CODE

```
import java.util.Scanner;

public class LongestSubstring {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String current = "";
        String longest = "";

        for (int i = 0; i < str.length(); i++) {
            String ch = str.substring(i, i + 1);
            int pos = current.indexOf(ch);

            if (pos != -1) {
                current = current.substring(pos + 1);
            }

            current = current + ch;

            if (current.length() > longest.length()) {
                longest = current;
            }
        }

        System.out.println("Longest substring: " + longest);
        System.out.println("Length: " + longest.length());

        sc.close();
    }
}
```

OUTPUT

```
Enter a string: abcabcbb  
Longest substring: abc  
Length: 3
```

```
Enter a string: pwwkew  
Longest substring: wke  
Length: 3
```

RESULT

The program successfully finds the longest substring without repeating characters using String operations such as `length()`, `substring()`, `indexOf()`, and String concatenation.